



Resume Annotator

PRESENTATION BY:

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Motivation:

- Building a resume for a Job description can be a tedious task.
- In this project we are addressing this critical issue and make it easier for Users, In Understanding mistakes and changes can be made to their resume.

We:

1. Created a Interface Using Streamlit where user has to upload resume and description of the job user is applying
2. Analyse the Resume and Job description
3. Provided ATS score of the resume w.r.t the Description with suggestions of analysis
4. Finally highlight the changes to be made in resume helping user decide where and what changes are to be made.
5. Mail resume to the user extracted from Resume



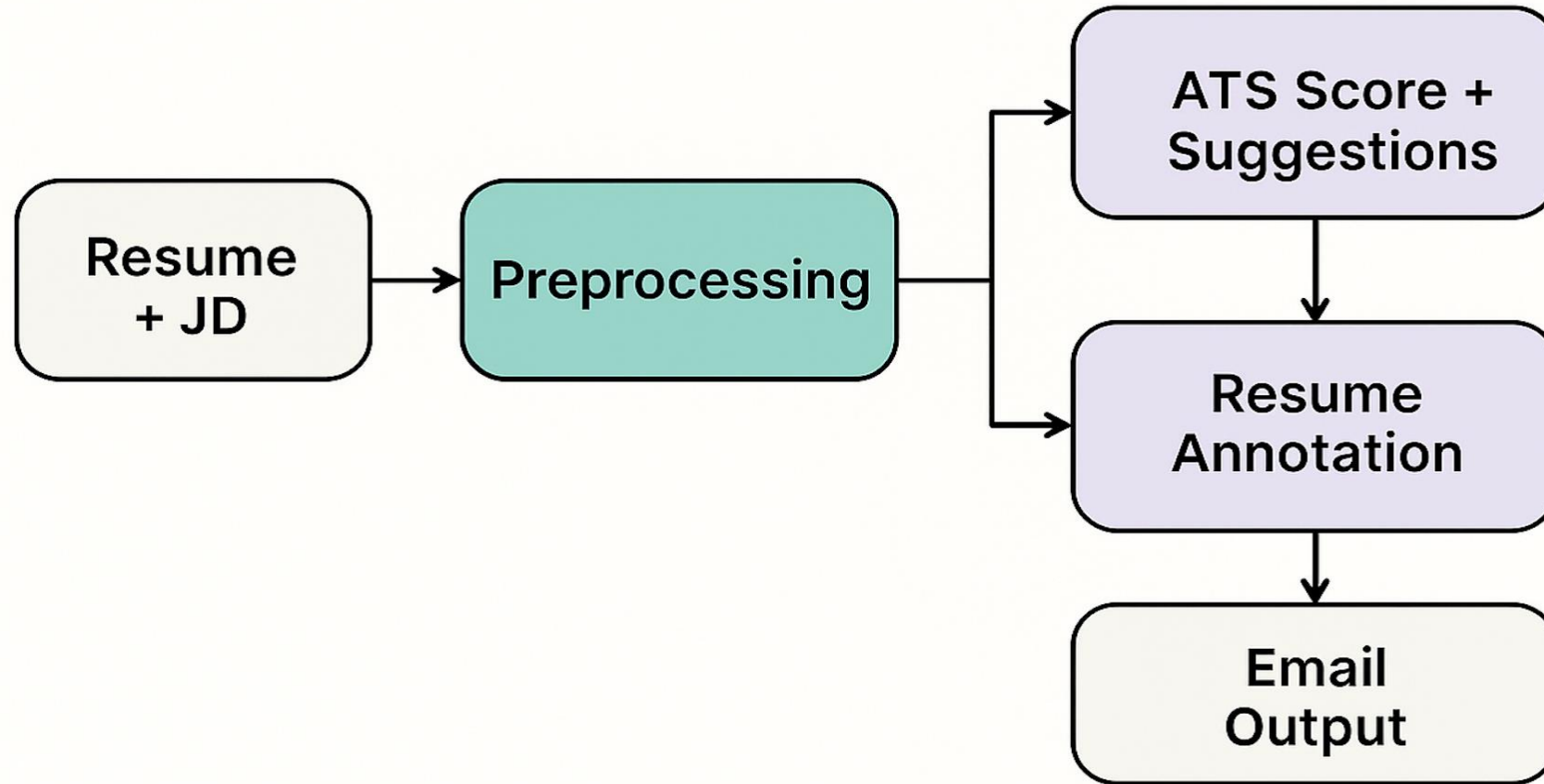
Sources

- Data collected from User Resume and Job Description.
- Custom Data on Rules, Restrictions, Guidelines, Examples for LLM
- Kaggle Resume Dataset

Packages and Models

- Streamlit, Mistral AI, GPT-4, BERT-xtremedistil-l6-h256-uncased, re, spacy, collections, Counter, transformers, sentence_transformers and More available at [Requirements](#)

Project Workflow



Preprocessing:

Here in the initial stages we did Text extraction from Job description and Resume

ATS score:

ATS Score is the sum of aggregated components such as

- Keyword Match Score
- Experience Alignment Score
- Competency & Skill Match Score
- Formatting & ATS Compliance Score
- Sentence Structure & Impact Analysis Score
- Job Description Mirroring Score

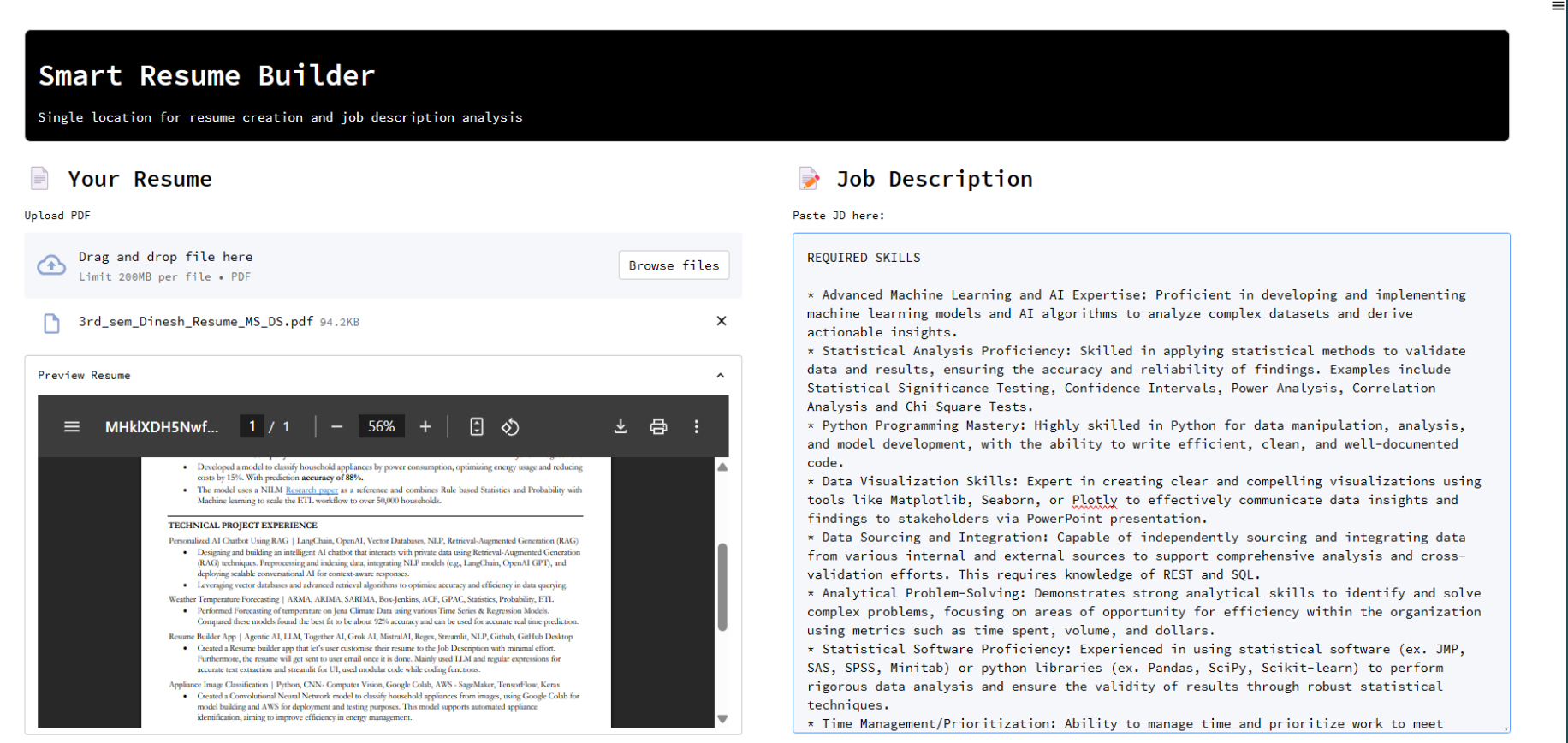
Resume Annotations:

- Red – Action verbs
- Green- metrics
- blue - Impact
- Hard Skills – orange
- Soft – yellow

Email notification:

We used Smtplib, Email to mail user his Annotated resume.

User Interface:



User uploads his Resume and relevant Job description in this landing page and clicks Analyse which does the preprocessing of the data.

UI Page-2



Page-2 has ATS analytics, Resume which gets Annotated and Side bar that has ATS metrics

app

Resume Builder

Match Rate



Upload & rescan

⚡ Power Edit

Searchability - 2 issues to fix

Hard Skills - 1 issue to fix

Soft Skills - 2 issues to fix

Recruiter Tips - 1 issue to fix

Formatting - 2 issues to fix

Smart Resume Builder

Single location for resume creation and job description analysis

ATS Analytics Job Description Resume

📌 Job Description Focus Analysis ↔

Label Frequency

• ACTION: 64 mentions

• HARD: 48 mentions

• SOFT: 28 mentions

• IMPACT: 19 mentions

• METRIC: 1 mentions

💡 Insight

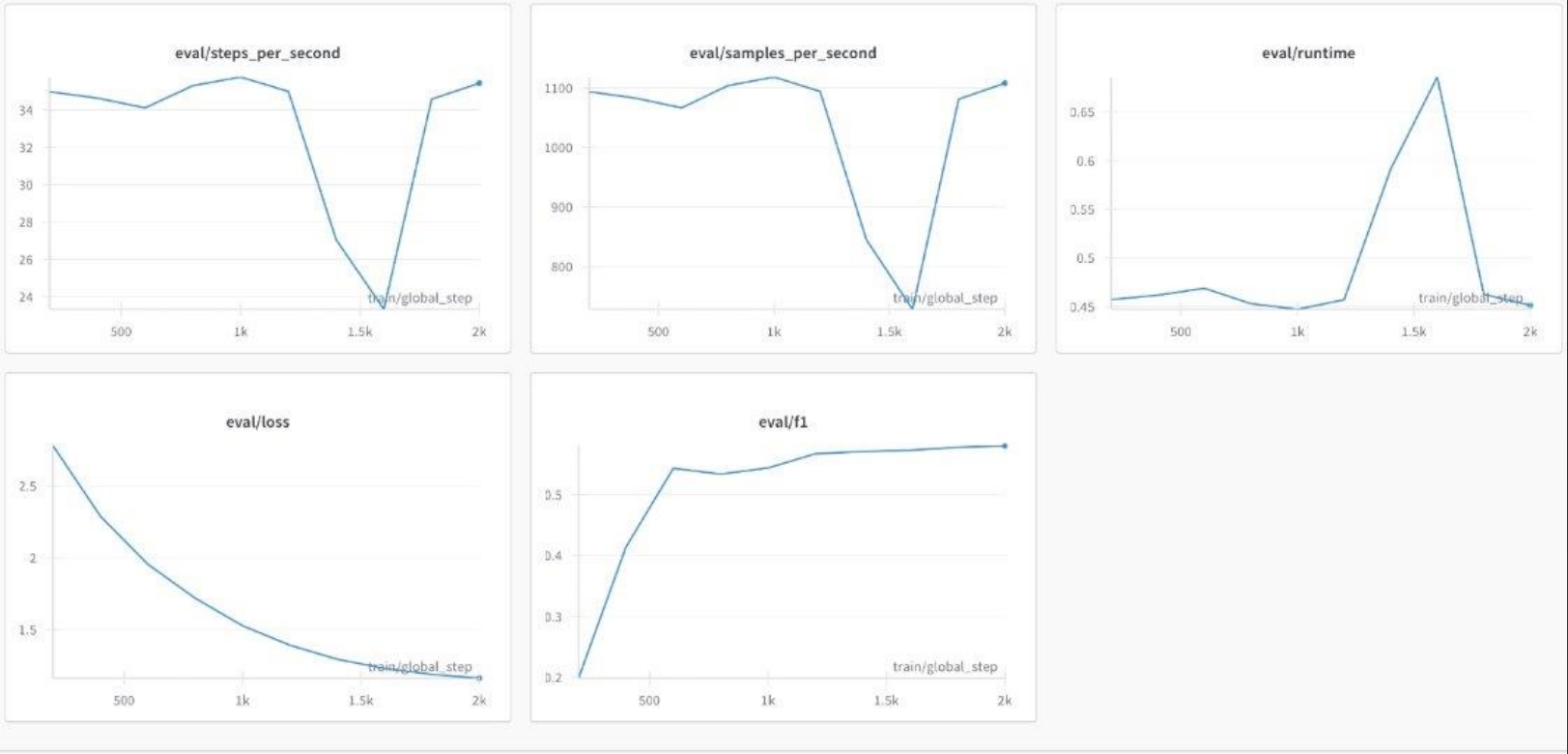
This job description emphasizes ACTION, HARD, SOFT.

👉 Make sure your resume uses strong action verbs to describe contributions, and highlights technical and platform-specific skills, and shows collaboration and communication abilities.



Bert Evaluation Metrics Used in Annotation:

"activation": "gelu",
"architectures":
 "DistilBertForTokenClassification",
"attention_dropout": 0.1,
"dim": 256,
"hidden_dim": 512,
"hidden_dropout": 0.1,
"initializer_range": 0.02,
"max_position_embeddings": 512,
"n_heads": 4,
"n_layers": 6,
"vocab_size": 30522,



Smart Resume Builder

Single location for resume creation and job description analysis

ATS Analytics Job Description Resume

resume.pdf

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EDUCATION

- George Washington University

Master of Science in Computer Science | GPA: 3.80

Dec. 2023 - May 2024

Graduate Teaching Assistant (CSCI 642) - Advanced DBMS II. Worked on Thrasos scripts to **reduce** infrastructure deployment for distributed systems and **improve** pipeline **efficiency** in **cloud** environments.

EXPERIENCE

- Advisia Solutions

Cloud Architect

Jan 2023 - Dec 2023

- TensorFlow**: Worked on implementing deep learning models using TensorFlow for structured and unstructured **data**. Focused on model optimization and deployment.
 - Apache Beam**: Built scalable **data** pipelines using Apache Beam for real-time and batch processing across Google Cloud Platform.
- VNB Consulting Services

Senior Software Engineer

Jun 2020 - Dec 2022

- Notifications**: Built **services** for email push and in-app notifications with **feature flag** driven **release** optimization, **tracking**, **monitoring**, and A/B testing.
 - Node.js**: Developed a **real-time** data processing and ingestion system from Redis to Cassandra with a REST API layer for serving offline **data**.
 - Workflows**: Used **Docker** (open-source) for building reusable and automated **data** pipelines, **improving** efficiency.
 - Data Collection**: Designed an internal crowdsourcing platform to collect and **manage** **data** across the **company**.
 - Dev Environment**: Set up Docker-based analytics environments on **cloud**, standardizing Python and R libraries used by **data** science teams.
 - Recommendations**: Contributed to new recommendation systems for Content's homepage and discovery, supported both training and real-time inference.
 - Content Discovery**: Improved search and browsing experience with sub-hourly personalization, enabling improvement and reducing optimization.

PROJECTS

- Credit Risk Prediction System:
 - Developed an ML pipeline using Logistic Regression and XGBoost to **predict** **credit** **risk** with 87% accuracy. Performed feature engineering and hyperparameter tuning, improving AUC-ROC by 12%. Contributed to core recommendation systems for Content's homepage and discovery. Supported both training and real-time inference.
 - Deployed model via Flask API and integrated into a dashboard for real-time decision making. **Applied** **recommendation** **techniques** to reduce monitoring and cut training time by 80%. Built **services** for email push and in-app notifications. Designed an internal crowdsourcing platform to collect and **manage** **data** across the **company**.
- Image Classification using Transfer Learning:
 - Fine-tuned ResNet50 and EfficientNet models to classify medical images with 92% validation accuracy. Set up Docker-based analytics environments on **cloud**, standardizing Python and R libraries used by **data** science teams. Built **services** for email push and in-app notifications with **feature flag** driven **release** optimization, **tracking**, **monitoring**, and A/B testing.
 - Used MLflow for experiment tracking and monitoring to **reduce** **hyperparameter** **tuning**. **Applied** **hyperparameter** **techniques** to reduce overfitting and cut training time by 50%.

SKILLS

- Machine Learning: TensorFlow, PyTorch, scikit-learn, XGBoost, LightGBM
- DBMS Engineering: SQL, Apache Spark, Pandas, Airflow
- Cloud & Deployment: AWS, Docker, Kubernetes, Flask
- Languages: Python, Java, C++

Results



 Resume Analysis
optimized for ATS



 Sent to user's
email



 User Resume
Annotated for Editing

References:

- Streamlit. (n.d.). *Streamlit documentation*. <https://streamlit.io>
- Scikit-learn Developers. (n.d.). *TF-IDF vectorizer — scikit-learn 1.4.1 documentation*. <https://scikit-learn.org>
- Bird, S., Klein, E., & Loper, E. (n.d.). *Natural Language Toolkit (NLTK)*. <https://www.nltk.org>
- Together AI. (n.d.). *Together AI documentation*. <https://docs.together.ai>

Thank you.

